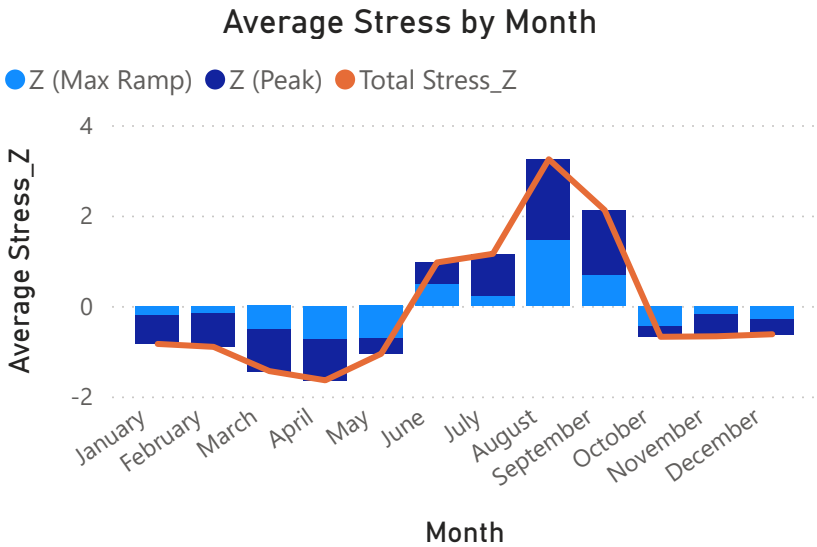
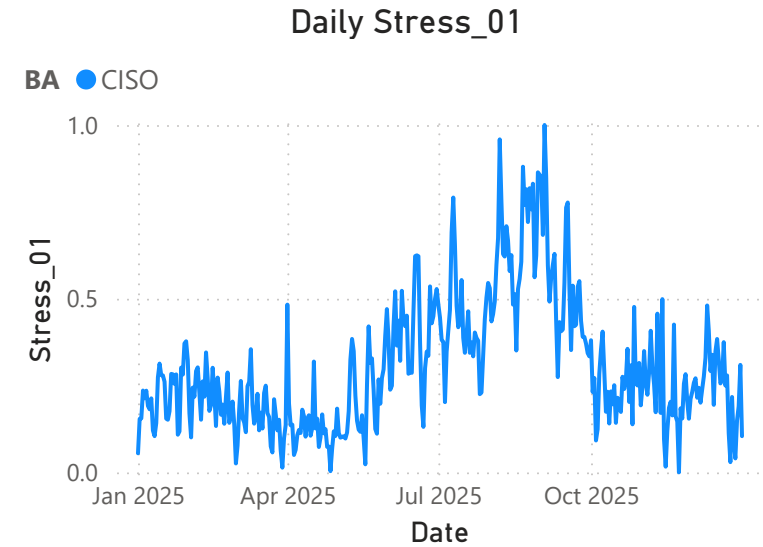


Electric Grid Stress Analysis - Overview

This dashboard visualizes a machine learning framework built on public grid data classifying daily stress regimes across multiple Balancing Authorities (BAs) using z-scores

Analytic Structure

Peak = Daily max power demand
Max_Ramp: Daily max power ramp between consecutive hours
 $Stress_Z = Z_Peak + Z_Max_Ramp$ (mean = 0, value $\approx$ std dev)
 $Stress_01 = \text{normalized } Stress_Z \text{ by BA}$ (min = 0, max = 1)



Balancing Authority (BA)

Select all

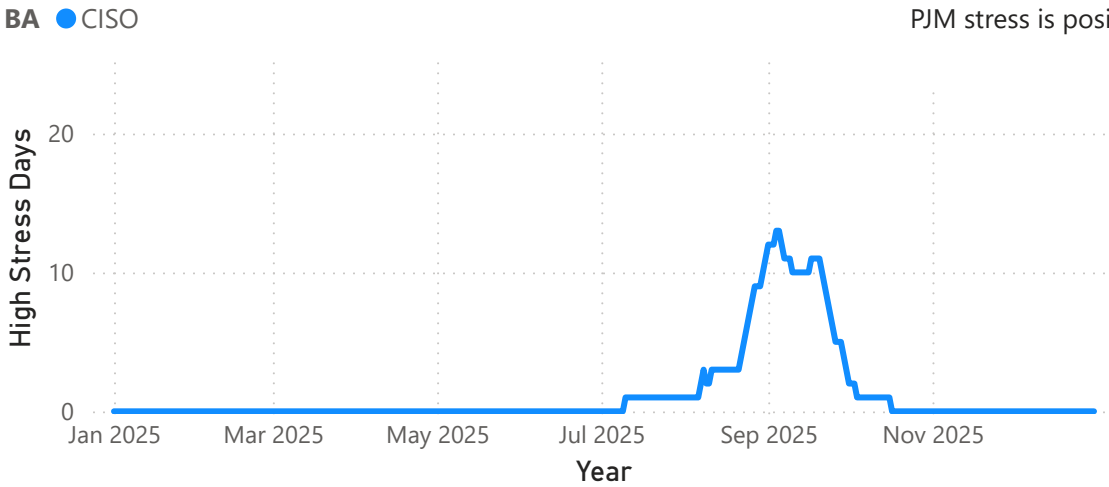
CISO

PJM

Most Stressful Days Table

BA	Stress_01	Date
CISO	0.82	August 23
CISO	0.82	August 25
CISO	0.83	August 27

Number of High Stress Days in the Previous 30 Days



Note: Stress by Month only informative by separated BAs
PJM stress is positive in winter months

Stress_Z (Max) Stress_01 (Avg) Most Stressed Day High Stress Days

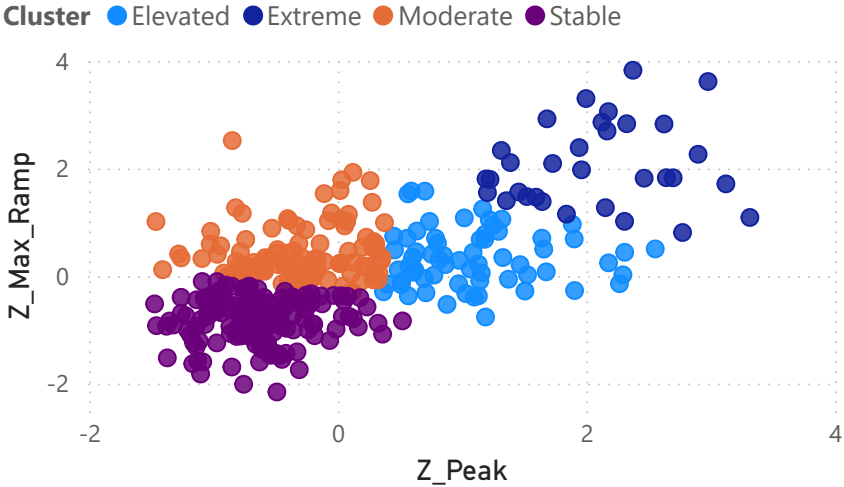
CISO	CISO	CISO	CISO
6.60	0.31	Sep 03	18

Both systems peak in summer, but CAISO's stress concentrates into a tight Aug-Sep window while PJM builds earlier and holds longer

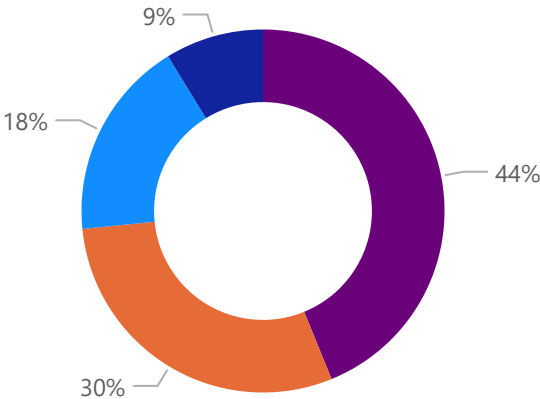
CISO has higher stress days but PJM has more high stressed days
• High Stress Days Threshold: $Z_01 > 0.7$

Electric Grid Stress Analysis - ML Clustering

Z-Score Clustering



Frequency by Cluster



Balancing Authority

CISO

PJM

Operators can use this calculator to determine how stressed their grid would be given specified conditions

Daily Stress Calculator

Peak Input (MW)

40000

Max Ramp Input (MW)

3000

Extreme

Identified Cluster

94%

Stress Percentile

3.39

Input Stress Z

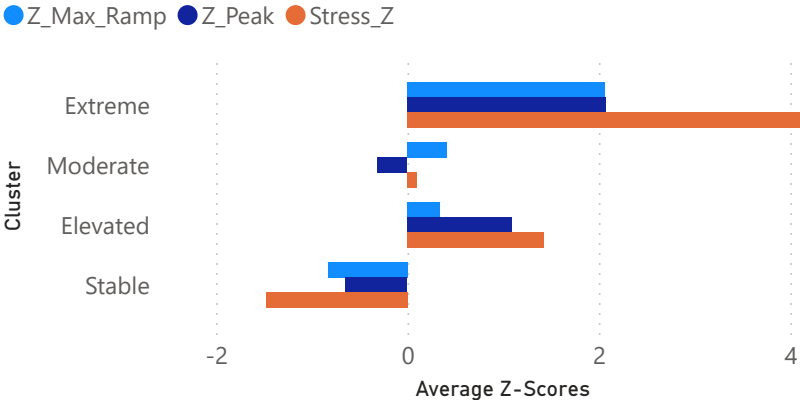
4.13

Cluster Mean Stress Z

Both BAs cluster similarly with instability increasing along the diagonal. CISO reaches higher extremes with more spread out datapoints

PJM contains more than double the extreme days
CISO contains nearly double the stable days

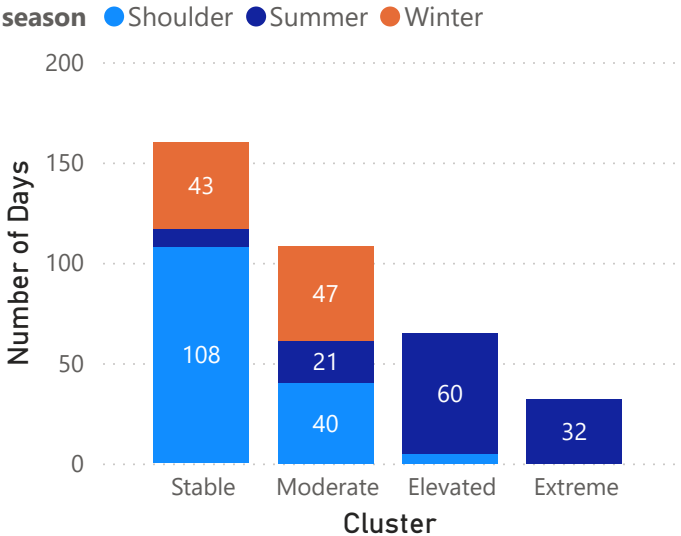
Average Z-Scores by Cluster



CISO's Extreme cluster combines peak and ramp stress nearly equally

- both must be elevated simultaneously. PJM's Extreme cluster is more peak-led (z_peak 1.58 vs z_ramp 1.12).

Number of Days by Cluster and Season



PJM has more extreme days (including winter days)

Approximate 2025 Ranges (MW)

CISO Peak: 23,000 - 45,000 || CISO Ramp: 1,000 - 5,000

PJM Peak: 78,000 - 160,000 || PJM Ramp: 2,500 - 9,100

k	CISO	PJM	Total
2	0.51	0.42	0.93
3	0.40	0.41	0.81
4	0.40	0.41	0.81
5	0.38	0.39	0.76
6	0.37	0.41	0.79
Total	2.06	2.04	4.09

Model Validation - Silhouette Scores

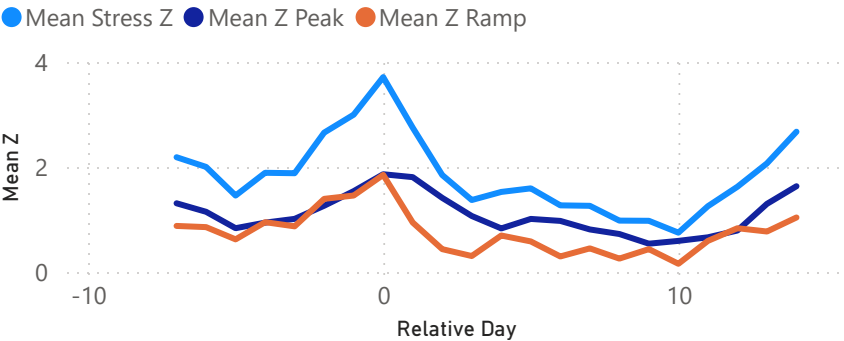
k=4 best for operational interpretability

- four regimes map well to Stable → Extreme
- k=3 collapses the moderate range;

k=6 adds granularity without adding insight

Electric Grid Stress Analysis - Event Drilldown

Composite Extreme Event (CAISO Summer)

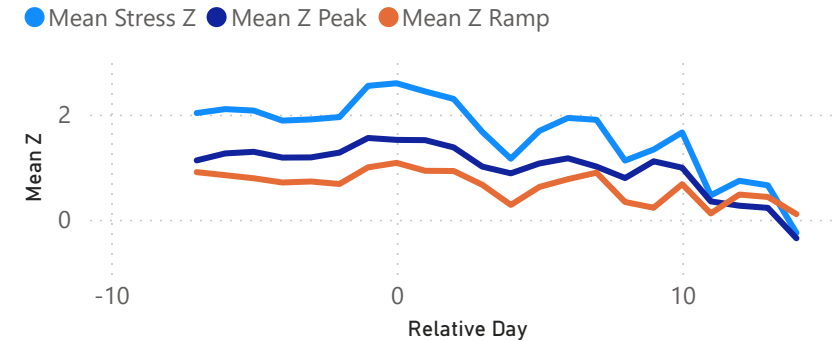


Ramp-amplified— solar drop-off drives the evening stress spike

- sharp ramp drives stress on extreme day when solar drop-off occurs

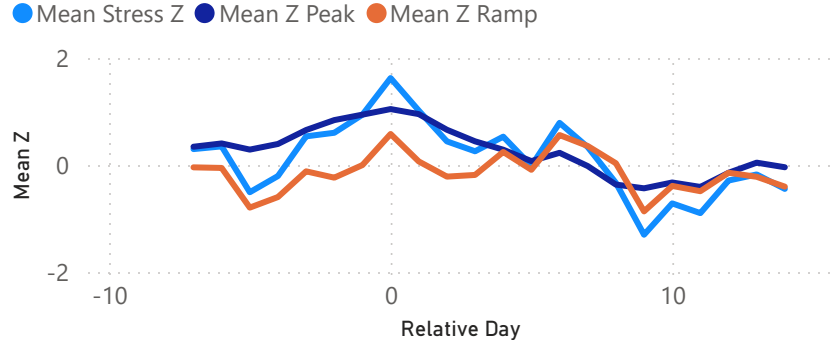
Composite events - An average of the extreme stress events. How stress builds and resolves around extreme events (day 0 = stress peak)

Composite Extreme Event (PJM Summer)



Load-dominated - slower build-up and drop off due to heat dome effect

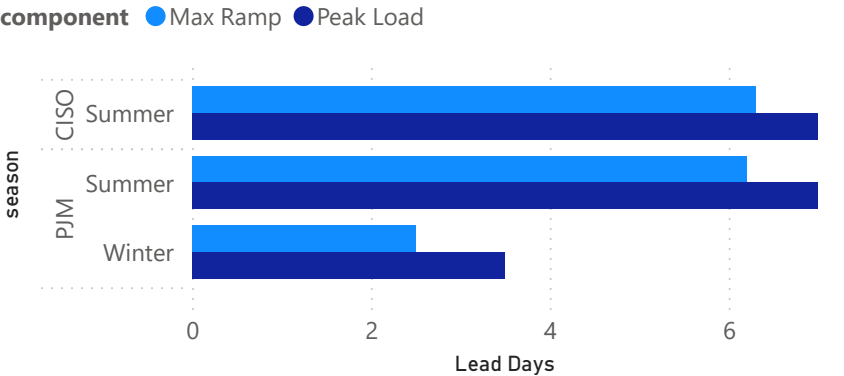
Composite Extreme Event (PJM Winter)



Load-dominated — quick cold snap raises demand

- ramp signal is weak or negative

Lead Days to Extreme Event

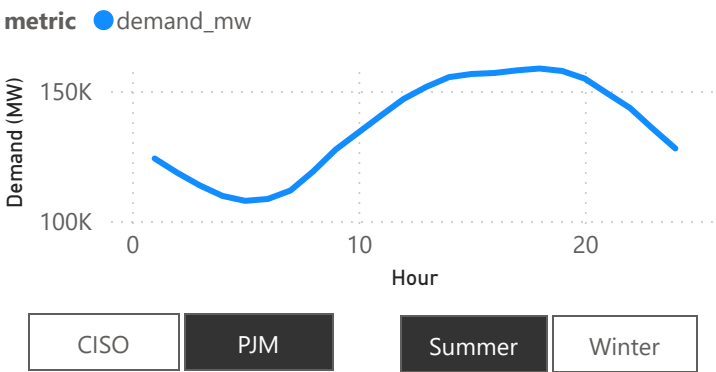


Above: Summer events give much more warning than winter PJM events

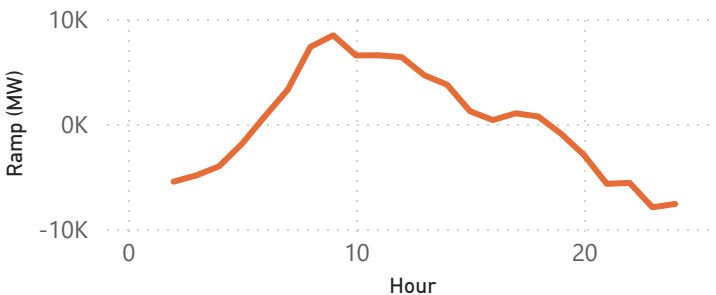
Right - Hourly differences by BA and season demonstrated

- i.e. CISO summer duck curve and PJM winter heating ramps
- CISO solar penetration is ~25% but PJM solar penetration is only ~5%

Hourly Demand on Extreme Day



Hourly Ramp on Extreme Day



Summary notes:

- CAISO stress is acute and ramp-amplified — few days, very high peaks. PJM stress is broader and load-led — more days, sustained demand. Different mechanisms require different responses.
- Compound event — heat wave elevates load for days; solar drop-off then triggers the ramp spike that pushes to Extreme"
- The additive stress formula can produce near-zero stress scores for high peak / low ramp days.
- In practice, such days are rare ($r=0.65$ between components), but the formula is not designed to flag isolated capacity-only stress.
- All data is based on 2025, but analysis can be applied to other years.